Electromagnetic induction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why changing magnetic flux can induce an emf.

SUCCESS CRITERIA - I can...

1. Identify a change in magnetic flux.
2. Predict whether a stationary arrangement produces sustained induction.
3. Use Lenz's law to describe opposition to change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **3 W** device runs for **5 s**. **Find transferred energy**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Phi = BA$; $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$.
Uniform perpendicular field for $\Phi = BA$. N is coil turns; the induced effect opposes the flux change.

Worked reasoning: A changing magnetic flux through a coil can induce voltage. Holding a magnet stationary relative to a fixed coil gives no induction from relative motion in this simple setup.

Your turn: Draw a magnet approaching a coil and **mark the changing field through** the loop. Contrast it with a stationary case.

Label field direction, loop area and what changes.

Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Compare a magnet held still** beside a fixed coil with a magnet moving into the coil.
2. Which can produce a changing flux?

Label field direction, loop area and what changes.

Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

04 Apply the physics

A lab group pushes a magnet into a fixed coil and records a positive meter deflection.
1. **Draw the approach** and withdrawal motions.
2. **Predict the sign** when the magnet is pulled back out along the same path without changing connections.

Label field direction, loop area and what changes.

Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why does Lenz’s law oppose the change in flux rather than always oppose the field itself?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Keep charge separation fixed and multiply one charge magnitude by **4**. **Compare electric-force magnitude**. **Label charges** and both interaction forces.



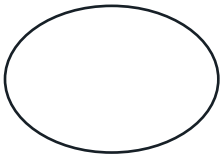
Charge A Charge B

Marzano check - circle: 1 2 3

07 Exit check

Name two ways to increase the magnitude of induced emf in a simple magnet-and-coil trial.

Label field direction, loop area and what changes.



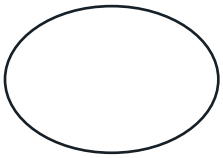
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Predict how faster approach** and more turns can change induced voltage.
2. **State which comparisons must keep** everything else in the setup fixed.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3